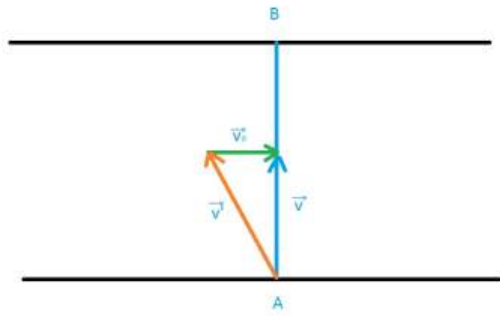


Question 1.8 : Two swimmers leave point A on one bank of the river to reach point B lying right across on the other bank. One of them crosses the river along the straight line AB while the other swims at right angles to the stream and then walks the distance that he has been carried away by the stream to get to point B. What was the velocity u of his walking if both swimmers reached the destination simultaneously? The stream velocity $v_o = 2.0$ km/hour and the velocity v' of each swimmer with respect to water equals 2.5 km per hour.

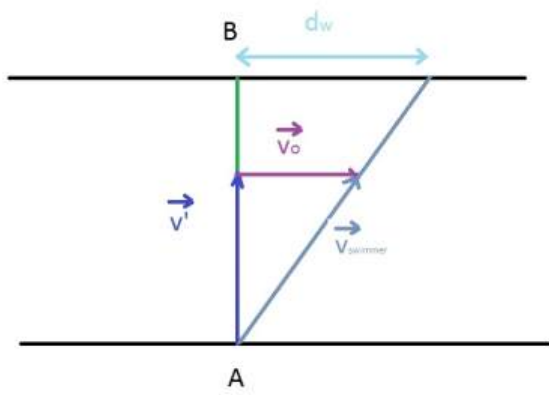
Solution Case I : Swimmer swims with final velocity along AB, this case is also called the Shortest Path case.



We see that $\vec{v}' + \vec{v}_o = \vec{v}$ and they form a right triangle
 $\Rightarrow v = \sqrt{v'^2 - v_o^2}$

time to reach point B = $\frac{AB}{\sqrt{v'^2 - v_o^2}}$

Case II : Swimmer swims at right angle to the stream, this would require the Shortest Time in crossing the river but the problem requires that he would have to walk to get to B.



We see that $\vec{v}' + \vec{v}_o = \vec{v}$ and they form a right triangle, though this time with $\vec{v}$ as the hypotenuse. It should be noted however that this time v' , v_o and v vectors are different from Case I while the magnitudes of v' and v_o are the same. Our problem Case II is independent in this regard from case I and we can choose the same names without loss of generality

Proceeding to solve the question

$$\frac{AB}{v'} = \frac{d_w}{v_o} \text{ (By similarity of triangles, from figure)}$$

$$\Rightarrow d_w = \frac{v_o}{v'} \cdot AB$$

Time t_1 to cross the river and reach the point (say C)

$$t_1 = \frac{AB}{v'}$$

Time t_2 to cross the distance d_w back to B

$$t_2 = \frac{d_w}{u}$$

$$\text{Total time} = t_1 + t_2 = \frac{AB}{v'} + \frac{d_w}{u} = \frac{AB}{v'} + \frac{v_o \cdot AB}{v' \cdot u}$$

Proceeding to solve the problem by comparing both the cases,

$$\frac{AB}{\sqrt{v'^2 - v_o^2}} = \frac{AB}{v'} + \frac{v_o \cdot AB}{v' \cdot u}$$

Solving further, we get the required form, first cancelling AB and cross multiplying v'

$$\frac{v'}{\sqrt{v'^2 - v_o^2}} - 1 = \frac{v_o}{u}$$

$$\Rightarrow u = \frac{v_o}{\frac{v'}{\sqrt{v'^2 - v_o^2}} - 1}$$

Calculating the value, $u = 3$ km/hr